Course Code: CSE3722	**Course Name:** IoT Networks, Architectures and Applications	
Credits: 03 (L-D-P: 3-0-0)	**Contact Sessions:** 03	
Batch: CSE 2023, 6th Semester **Academic Year:** 2025-26	**Semester Duration:** 19 Jan 2026 to 08 May 2026	
Course Faculty: Dr. Sandeep Kumar *Contact details: sandeep.kumar@bmu.edu.in, 9133872511, Office No. 67, E2 4th floor* Dr. Abhishek Jain *Contact details: abhishek.jain@bmu.edu.in, 9466852190, Office No. 24, E2 4th floor* Dr. Sumit Badotra *Contact details:* Sumit.badotra@bmu.edu.in *9816338848, Office No.3, 2nd floor, Gateway-B*	**Course Coordinator:** Dr. Sandeep Kumar Email: sandeep.kumar@bmu.edu.in Contact No: +91-9133872511 Office: 67, IV Floor, E2 Building	

Aim of the course: This course is designed to provide students with solid technical knowledge and skills to understand the Internet of Things (IoT) systems. The beginning portion focuses on the basic concepts, architectures, applications and design challenges of the internet of things. The various topics on the IoT protocols, gateway, computing layers, security and wireless sensing have been covered in the middle portion of this course. The remaining part takes up the IoT circuits, embedded systems, case studies and a number of online IoT platforms.

Course Outcomes (CO): At the end of the course the students should be able to do the following:

	Course Outcome Statement	Mapped Component	Associated Activities / Assessment
CO1	Understand the basic concepts, network architectures, applications and design challenges in the internet of things.	Employability	→ Intensive theory sessions. → In-class tool demonstrations → Industry case studies etc.
CO2	Analyse the functionality of various computing paradigms, sensor networks, IoT protocols and security aspects.	Skill Development	→ Hands-on practical sessions in labs. → Experiential learning on the simulation platforms e.g. WokWi, Proteus, CupCarbon, MATLAB Simulink etc. → Data communication to the Android Apps and cloud platforms.
CO3	Apply the acquired knowledge to design IoT circuits and systems.	Entrepreneurship, Skill Development	→ Getting familiar with cloud services such as Blynk, ThingSpeak, HiveMQ, CloudMQTT, Bolt, PushBullet etc. → Design and development of the prototypes based on the real world applications.

Prerequisite: if any

1. Basic knowledge of microcontroller boards.
2. Programming in Python and C.

Topics of the course:

Topics	Number of Sessions (48)	Course Outcomes	Reference
Basics of networking, OSI and TCP/IP models, Addressing, TCP connections.	2 Theory + 1 Lab	CO1	T1: Chapter-1
What is IoT and why is IoT.	3 Theory + 1 Lab	CO1	T1: Chapter-4
Applications of IoT, potential and challenges.	1 Theory + 1 Lab	CO1	T1: Chapter-15
IoT network architectures.	3 Theory + 1 Lab	CO1	R1: Chapter-2
Types of communication: M2M, machine to cloud, etc.	2 Theory + 1 Lab	CO2	T2: Chapter-1 and 3.
Network protocols and standards for IoT Connectivity: IEEE-802.15.4, ZigBee, RFID, NFC, Z-Wave, SigFox, LoRa, NB-IoT, Wi-Fi, Bluetooth.	3 Theory + 1 Lab	CO2	T1: Chapter-7
Network protocols and standards for IoT communications: IPv6, RPL, 6LoWPAN, MQTT, MQTT-SN, CoAP, AMQP, XMPP, WebSocket, EPC and uCode.	3 Theory + 1 Lab	CO2	T1: Chapter-8.
IoT Gateway: Overview of hardware and software. Computing Technologies: Cloud and fog computing.	3 Theory + 1 Lab	CO2	T1: Chapter-10, 11 and Class Notes.
Basics of Security: Network, transport and application layer Security. Wireless Sensor Networks: Nodes, features, architecture and deployment.	2 Theory + 1 Lab	CO2	T1: Chapter-2 and 3.
Circuits: overview of circuits used in IoT, battery current, and wireless links, digital computing and analog to digital interfaces.	3 Theory + 2 Lab	CO3	Class Notes
Embedded systems: internet connectivity and MGC architecture, Cortex M and BLE.	3 Theory + 1 Lab	CO3	Class Notes
IoT case studies: Industrial IoT, home automation, internet of vehicles, e-Health, smart cities, etc.	3 Theory + 2 Lab	CO3	R2: Chapter-3, and Class Notes.
Open source IoT platforms: CupCarbon, Wokwi, Proteus, PushBullet, Bolt, Blynk, ThingSpeak etc.	1 Theory + 2 Lab	CO3	Class Notes

CO/PO Mapping: (1: weak, 2: medium, 3: strong)

CO/PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1			2		2							1	2		2	
CO2		3									1				2	
CO3	3		3	2				1		2	2			1		3

Learning Resources:

Textbooks:

(T1): "Introduction to IoT" by Sudip Misra, Anandarup Mukherjee and Arijit Roy. Cambridge University Press, 2021 edition.

(T2): "Internet of Things: A Hands-on Approach" by Arshdeep Bahga and Vijay Madisetti. Universities Press India Pvt. Ltd., 1st edition (2016).

Reference Book:

(R1): "IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things" by David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton and Jerome Henry. Cisco Press, 2017 Cisco Systems, Inc.

(R2): "Building The Internet of Things with IPv6 and MIPv6: The Evolving World of M2M Communications" by Daniel Minoli. Wiley, 2013 by John Wiley & Sons, Inc.

Assessment Pattern: The final grade will be based on the marks/ grades obtained in the mid-term and end-term evaluations along with other components as defined in the evaluation plan/table. The relative grading method defined in the academic regulations of the university will be followed to grade the students. The student has to secure a minimum of 40% of marks after completing all the assessments in the following table to become eligible for grading.

Evaluation Component	**Weightage (%)**	**Evaluation Schedule/Timeline**	**Evaluation Type/Remarks**	**CO mapping**
Mid-Term Exam	30%	As per academic calendar	Closed book	CO1, CO2
Lab Assignments	20%	Continuous	Based on lab experiments	CO1, CO2, CO3
Coursera Certification	10%	To be completed by 2nd/3rd week of April	Closed book	CO1, CO2, CO3
End-Term **(Project-Based)**	40%	1st week of May	Demonstration, viva and report	CO1, CO2, CO3

Experiential Learning Component:
The lab assignments and end-term project will facilitate 60% experiential learning.

Student Responsibilities:

- To attend lectures and labs in order to obtain all the course material.
- To check announcements on G-Class and emails on a regular basis.
- To submit assignments on time.
- To regularly, check the marks on Maitri portal and make sure that they are up to date.

Attendance Policy: Students are expected to attend the classes regularly. Failure to attend the classes may dip the attendance below expected percentage that will result in reduction of grade or course drop-out as per the University's policy.

Assignments: Each assignment will have a deadline for submission and every student should submit his/her assignment on or before that timeline.

Recourse Examination Policy: In case a student fails the course, a one-time recourse is permitted as per the academic regulations of the University. Recourse is allowed only for the End Semester examination.

Make-up policy: No make-up work will be given for unexcused absences. The faculty needs to be informed in advance in case the student is not going to be able to submit an assignment or take any evolution component, and it is at the discretion of the faculty to sanction make up for an evaluation component.

Behavior Expectations: No mobile phones and other distractive gadgets are permitted in the class.

Academic Dishonesty/Cheating/Plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.